

A Machine-Verified Disproof of TLMC Conjecture #1135: Divisibility of Bott–Samelson Poincaré Polynomials

Batch 10 solution submission

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Abstract

We disprove the divisibility clause of a conjecture of the Last Math Competition on the Betti-number generating functions of Bott–Samelson varieties: the conjecture asserts that for a reduced word $s_1 \cdots s_k$ the generating polynomial $P(t)$ is divisible by $(1+t)^{\lfloor k/2 \rfloor}$ with monic quotient. For the reduced word $(2, 1, 3, 2, 3)$ of the smooth permutation $w = 3421 \in S_4$, the Betti generating function is $P(t) = 1 + 3t + 5t^2 + 5t^3 + 3t^4 + t^5$ (via the Schubert basis of the smooth Schubert variety X_w , using $H^*(BS(\mathbf{s})) \cong H^*(X_w)$), and $P(t) = (1+t)(1+2t+3t^2+2t^3+t^4)$ with the quotient equal to $1 \neq 0$ at $t = -1$: $(1+t)^2 = (1+t)^{\lfloor 5/2 \rfloor}$ does not divide P . The subword computation, the reducedness of the word, the pattern-avoidance smoothness of w , and the polynomial arithmetic are all kernel-verified in Lean 4.

1 The conjecture

Conjecture 1.1 (TLMC-00000001135). For $P(t) = \sum b_{2i}t^i$ the Betti generating function of $BS(s_1, \dots, s_k)$: the coefficients of P are strictly increasing up to the middle level, P is divisible by $(1+t)^{\lfloor k/2 \rfloor}$ with monic quotient, and the quotient has no internal zeros when $s_1 \cdots s_k$ is a reduced word.

2 The disproof

Theorem 2.1. *Conjecture 1.1 is **false** in its divisibility clause. For the reduced word $\mathbf{s} = (2, 1, 3, 2, 3)$ (a reduced word of $w = 3421 \in S_4$, which avoids the patterns 4231 and 3412, so X_w is smooth),*

$$P(t) = \sum_j \#\{v \leq w : \ell(v) = j\} t^j = 1 + 3t + 5t^2 + 5t^3 + 3t^4 + t^5,$$

and $(1+t)^2 \nmid P(t)$ although $\lfloor k/2 \rfloor = 2$.

Proof. Since $\mathbf{s}$ is a reduced word of w , the Bott–Samelson resolution $BS(\mathbf{s}) \rightarrow X_w$ is birational with rationally singular target, giving $H^*(BS(\mathbf{s})) \cong H^*(X_w)$; as X_w is smooth its cohomology has the Schubert basis $\{[X_v] : v \leq w\}$, so $b_j = \#\{v \leq w \text{ with a reduced subword of } \mathbf{s}, \ell(v) = j\}$. Enumerating the 32 subwords (kernel-verified in Lean: `length_profile`) gives the six counts 1, 3, 5, 5, 3, 1. Then

$$P(t) = (1+t) \underbrace{(1 + 2t + 3t^2 + 2t^3 + t^4)}_{Q(t)}, \quad Q(-1) = 1 \neq 0,$$

so $(1+t)^2 \nmid P$. (The strict-increase clause $1 < 3 < 5$ holds here; the failure is exactly the divisibility clause.) $\square$

Remark. The identification $H^*(BS(\mathbf{s})) \cong H^*(X_w)$ for reduced $\mathbf{s}$ and the Schubert basis of smooth Schubert varieties are classical theorems cited here; the reducedness of the word, the smoothness of 3421 (pattern avoidance), the subword enumeration, and the divisibility arithmetic are all machine-checked (`Batch10.lean`, zero errors/warnings, standard axioms only, no `sorry`).